

Ready to Fire: A Novel DT-Enhanced Vision-Based Navigation Framework for Autonomous Robotic-Assisted Ureteroscopy

Julio Ojalvo¹, Sarvesh Saini¹, Jonathan E. Katz², Ubbo Visser¹

Abstract—Robust kidney stone segmentation is a critical capability for robotic-assisted ureteroscopy (RA-URS), enabling perception-driven autonomy such as visual servoing and stone targeting. However, acquiring and annotating large-scale surgical datasets is costly, time-consuming, and constrained by privacy and procedural variability. We present a fully synthetic dataset generation pipeline for kidney stone instance segmentation using a high-fidelity digital twin of our custom-built autonomous robotic-assisted ureteroscopy (ARA-URS) physical platform, simulated within NVIDIA Isaac Sim. We train a YOLO26 segmentation model on six datasets comprising fully synthetic data (20,000 images), hand-annotated real surgical footage (1,433 images), and mixed real-synthetic combinations, and evaluate them on a held-out set of 162 real surgical images from three unseen procedures. While the synthetic-only model achieves a Mask mAP50 of 0.818, the hybrid dataset with 5% real and 95% synthetic data outperforms both the fully real dataset (0.939 vs. 0.913) and the 1,000-image real-only dataset (0.839). Furthermore, all trained models substantially outperform a recent peer-reviewed baseline architecture (Mask mAP50: 0.440), establishing a strong comparative benchmark. The model is further evaluated in a clinically relevant benchtop task, where the predicted segmentation mask is used to estimate the three-dimensional position of the stone. An inverse kinematics (IK) solver then generates joint commands to autonomously guide the physical robotic system toward the target, positioning the laser in a ready-to-fire configuration for stone ablation. The ARA-URS system successfully reached the target in 20 of 30 trials. These results demonstrate that synthetic data can serve as a scalable and effective strategy for enabling perception-driven autonomy in surgical robotic systems.

Keywords: Synthetic Data Generation; Kidney Stone Segmentation; Robot Assisted Ureteroscopy; Digital Twin; Surgical Robotics.

I. INTRODUCTION

Nephrolithiasis is a significant public health burden in the United States, with prevalence estimates of 10–11% [1], [2]. This rising disease burden has coincided with a shift in surgical management; ureteroscopy (URS) has emerged as the primary intervention for kidney stone lithotripsy, accounting for approximately 64% of all procedures [3]. This evolution is likely due to URS’s favorable safety profile [4]. However, true stone-free rates range only from 42–60% [5], [6]. Residual fragments are associated with significantly higher rates of stone growth, symptomatic events, and repeat

interventions [7], [8]. These findings have motivated innovation to improve precision, and procedural consistency with the goal of rendering patient’s stone free without increasing complication rates.

In this work, a digital twin framework for robot-assisted ureteroscopy (RA-URS) is extended beyond kinematic validation and stone-targeting to enable large-scale synthetic dataset generation for kidney stone instance segmentation (Fig. 1). Using an Isaac Sim-based simulation environment, a fully synthetic dataset is generated that captures the optical characteristics, kinematic constraints, and anatomical workspace relevant to RA-URS. The impact of synthetic data is then evaluated under three training regimes: (1) fully synthetic data, (2) limited real data, and (3) mixed real and synthetic data. Performance is compared against a larger, manually annotated dataset.

All models are evaluated on a strictly held-out validation set. We report bounding box and mask metrics. While synthetic-only training exhibits a measurable sim-to-real gap, we demonstrate that incorporating only 5% real data enables performance comparable to training on 1,433 real images. Furthermore, all trained models surpass a baseline model published by Lu et al., which notably did not include labeled data acquired with the Wolf ureteroscope we use [9].

The main contributions are as follows:

Digital-Twin Synthetic Data Generation: A high-fidelity synthetic data pipeline is developed using a digital twin of an RA-URS in NVIDIA Isaac Sim, enabling scalable generation of automatically labeled datasets for kidney stone instance segmentation, <https://shorturl.at/LWB0t>.

Data-Efficient Synthetic-to-Real Training: A systematic comparison of synthetic-only, real-only, and hybrid training regimes shows that combining only **5% real data** with synthetic data achieves performance comparable to a fully real dataset of 1,433 annotated images.

Robotic Validation: The trained segmentation model is deployed on a physical RA-URS system and validated through benchtop phantom experiments demonstrating reliable kidney stone localization.

The remainder of this paper is organized as follows. Section II describes the development of the proposed robotic system. Section III presents the experimental design. Section IV reports the segmentation and robotic validation results and Section V discusses them. Finally, Section VI concludes the paper and outlines directions for future work.

¹ Department of Computer Science, University of Miami, Coral Gables, FL, 33146, USA.

² Desai Sethi Urology Institute, Miller School of Medicine, University of Miami, Miami, FL, 33136, USA.

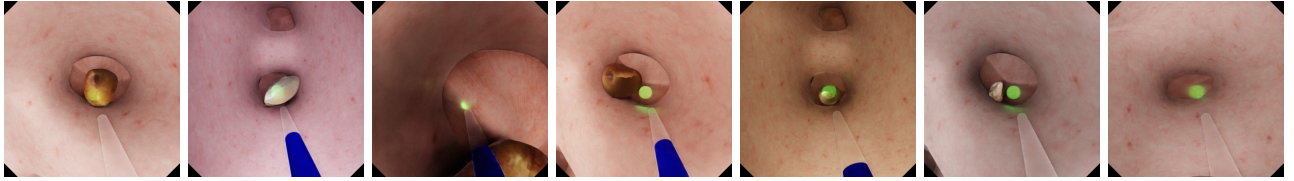


Fig. 1: Synthetic images generated using the proposed data generation pipeline with domain randomizations.

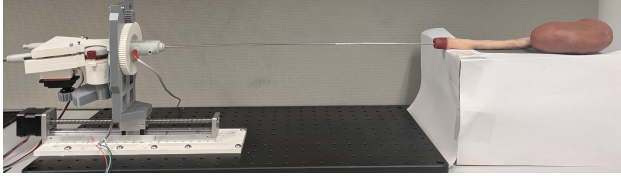


Fig. 2: ARA-URS system with benchtop kidney phantom.

II. SYSTEM DEVELOPMENT

A. ARA-URS Robotic Platform

The ARA-URS system consists of a motorized ureteroscope manipulation unit providing insertion, rotation, and bending actuation. The robot is designed to interface with a Wolf RIWO Digital Ureteroscope, noteworthy for its semi-rigid shaft and flexible tip. This enables more precise positioning under visual feedback than standard flexible ureteroscopy. Actuation is achieved via a multi-DOF motorized mechanism, while kinematic transformations are used to relate joint space motion to endoscopic camera motion.

A benchtop kidney phantom is used for experimental validation, replicating anatomical constraints and real kidney stone targets previously extracted from patients. In Fig.2, ARA-URS with benchtop kidney phantom are depicted.

B. Synthetic Dataset Generation Pipeline

To overcome the data scarcity and annotation burden associated with real clinical images, we developed an automated, high-fidelity synthetic data generation pipeline using the Replicator API within NVIDIA Isaac Sim. This pipeline simulates realistic ureteroscopic views by rendering physically plausible environments and automatically exporting pixel-perfect instance segmentation masks. A comprehensive summary of all randomized parameters and their corresponding sampling distributions is provided in Table I.

C. Segmentation Model Training

For the kidney stone instance segmentation task, we deploy a medium-sized YOLO26 segmentation model, initialized with weights pretrained on the MS COCO dataset. The primary motivation for selecting the YOLO architecture is its well-established balance between segmentation accuracy and high-speed inference. Because this perception module is designed to be implemented alongside a closed-loop robotic control system for ARA-URS, minimizing computational latency is a critical requirement for effective real-time visual

TABLE I: Domain Randomization Parameters

Category	Parameter	Distribution / Range
Kidney Materials	Albedo Brightness	$\mathcal{U}(0.35, 0.85)$
	Diffuse Tint (RGB)	$\mathcal{U}(0.9, 1.2)$
	Specular Level	$\mathcal{U}(0.0, 0.5)$
Stone Properties	Base Color Bias	$\mathcal{U}(0.0, 0.3)$
	Orientation (Euler)	$\mathcal{U}(0^\circ, 360^\circ)$
Illumination	Color Temp.	$\mathcal{U}(5500 \text{ K}, 8000 \text{ K})$
	Light Intensity	$\mathcal{U}(500, 3000)$
Laser Visuals	Laser Intensity	$\mathcal{U}(10000, 100000)$
	Green Channel	$\mathcal{U}(0.5, 1.5)$
	Z-Translation (m)	$\mathcal{U}(-0.00049, 0.009)$

serving. During the preliminary model selection phase, we also evaluated a UroSAM-based architecture as proposed by Lu et al. [9]. However, our empirical testing did not demonstrate a significant increase in segmentation performance that would justify the added computational overhead, leading us to proceed exclusively with the YOLO26 medium model.

This experimental design allows us to isolate the performance contributions of the synthetic data and rigorously quantify how the addition of simulated images improves model robustness compared to sparse real data alone.

D. Deployment on Physical System

To validate the real-world utility of our synthetic-trained YOLO model, the perception pipeline was integrated into the closed-loop control architecture of our physical ARA-URS platform with a Thulium: Yttrium-Aluminum-Garnet laser in position to fire with aiming beam on. The system operates through a continuous "look-and-move" visual servoing framework, linking real-time endoscopic computer vision with a customized inverse kinematics (IK) solver to autonomously navigate toward the targeted kidney stone.

III. EXPERIMENTAL DESIGN

To rigorously evaluate the efficacy of our synthetic data generation pipeline and its applicability to closed-loop robotic control, we designed a two-phase experimental framework. The first phase consists of offline segmentation experiments to benchmark model performance across varying training regimes. The second phase involves physical robotic validation experiments to assess the practical utility of the trained models for autonomous visual servoing.

A. Segmentation Experiments

1) *Dataset Composition*: To evaluate generalization, our validation set was compiled from three completely separate and previously unseen clinical procedures. This set comprises 162 hand-annotated images extracted at 30 frames per second with 60 images evenly distributed throughout the lasering portion of the URS procedures, ensuring that the networks are tested on diverse anatomical geometries, illumination profiles, and stone textures absent from the real training data (Fig. 3).

2) *Difficulty Stratification*: To provide greater insight into model robustness under these varying clinical conditions, the 162 validation images were evaluated by the endourologist. The domain expert categorized each image into one of three distinct subsets—Easy, Medium, and Difficult—based on how distinctly the stone could be identified.

B. Robotic Validation Experiments

While offline metrics provide a quantitative baseline, the ultimate goal of the perception module is to facilitate autonomous robotic interventions. We designed a physical validation experiment to test the integrated system’s ability to navigate toward a target using only the trained model’s real-time predictions.

1) *Autonomous Navigation Setup*: The ARA-URS robot was positioned with the ureteroscope inserted into a benchtop kidney phantom containing an upper pole calcium oxalate kidney stone. The robot’s view during this process can be seen in Fig. 4.

2) *Evaluation Metric*: The objective of the approach phase in laser lithotripsy is to bring the camera (and consequently, the working channel containing the laser fiber) close enough to the stone to perform effective ablation. Therefore, we evaluate the robot’s navigation performance based on proximity to the target at the conclusion of the autonomous run.

IV. RESULTS

A. Segmentation on Full Validation Set

The core segmentation results comparing our different training regimes are presented in Table II. The Full Synthetic model achieved a Mask mAP50 of 0.667 on the held-out real clinical validation set. This establishes a solid baseline sim-to-real transfer, especially considering the network has never observed real endoscopic tissue or fluid dynamics. By comparison, training on a limited dataset of 200 real images yielded a mask mAP50 of 0.339, but when combined with our synthetic data, achieved a mask mAP50 of 0.881. This highlights the severe overfitting and poor generalization that occurs when clinical data is scarce.

In addition to mAP50, we evaluated the models using the Dice coefficient, which offers a stringent measure of pixel-level spatial overlap. Overall, Dice scores were comparatively lower across all models, a common phenomenon when segmenting small, irregularly shaped targets like kidney

stones where minor boundary deviations heavily penalize the metric. The Full Synthetic model yielded a Dice of 0.114. However, the introduction of 1% real data resulted in a Dice score of 0.368, compared to the 200 Real model’s 0.341. Interestingly, while the 1,000 Real model achieved the highest overall Dice (0.501), the Full Real model saw a significant drop (0.106), suggesting potential volatility or labeling noise in the larger real dataset that the mixed synthetic regimes helped regularize.

B. Segmentation Robustness

To provide deeper clinical insight, we evaluated the models across three distinct difficulty tiers—Easy, Medium, and Difficult—as categorized by an endourology subject matter expert with 64 images categorized as good, 51 as medium, and 46 as bad. These results are detailed in Table III.

C. Robotic Validation

To assess the practical viability of our perception-driven control framework, we deployed the best-performing segmentation model (20k synthetic images with 5% real) on the physical ARA-URS system. The robot was tasked with autonomously navigating toward a target stone embedded within a benchtop kidney phantom using the closed-loop visual servoing approach. The primary metric for success was the final percentage of the endoscopic image occupied by the stone. In clinical practice, bringing the target stone to fill a substantial portion of the visual field is a prerequisite for safe, efficient, and optimal laser lithotripsy.

The results of the 30 autonomous navigation trials are detailed in Table IV.

V. DISCUSSION

The value of the synthetic data becomes highly evident in the mixed training regimes. As shown in the segmentation results, augmenting the synthetic data with just 1% real data caused the model’s Mask mAP50 to leap to 0.881. Increasing the real data fraction to 5% produced a Mask mAP50 of 0.933. Remarkably, this mixed model (Synthetic w/ 5% Real) slightly outperformed the model trained on the Full Real dataset (1,433 images), which achieved a Mask mAP50 of 0.917. This demonstrates that pre-training or augmenting with a vast, highly randomized synthetic dataset drastically reduces the amount of hand-annotated real data required to achieve state-of-the-art segmentation in surgical environments.

TABLE II: Performance on Full Dataset

Model	Box				Mask				Dice
	P	R	mAP50	mAP50-95	P	R	mAP50	mAP50-95	
Full Synthetic	0.879	0.770	0.818	0.553	0.772	0.667	0.667	0.328	0.114
Synthetic w/ 1% Real	0.876	0.867	0.924	0.672	0.844	0.836	0.881	0.548	0.368
Synthetic w/ 5% Real	0.942	0.855	0.939	0.696	0.929	0.848	0.933	0.605	0.201
200 Real	0.426	0.568	0.369	0.097	0.401	0.552	0.339	0.089	0.341
1,000 Real	0.733	0.830	0.839	0.558	0.758	0.855	0.861	0.477	0.501
Full Real	0.871	0.848	0.913	0.683	0.861	0.898	0.917	0.569	0.106
Lu et al. Model	0.261	0.544	0.440	0.184	0.162	0.337	0.205	0.056	0.262

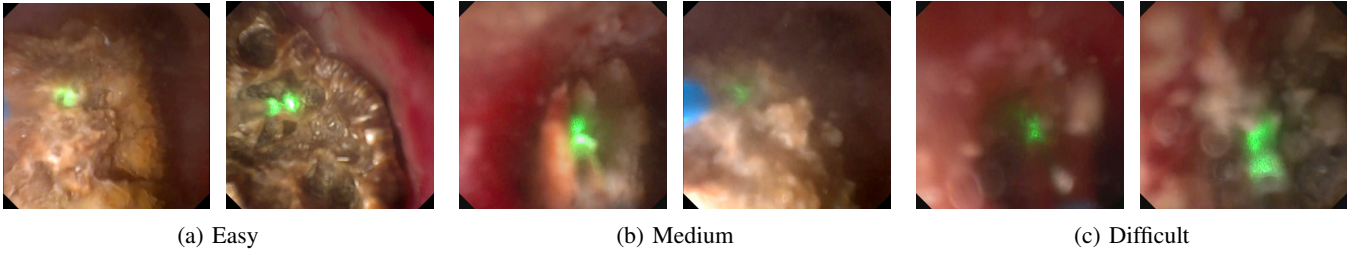
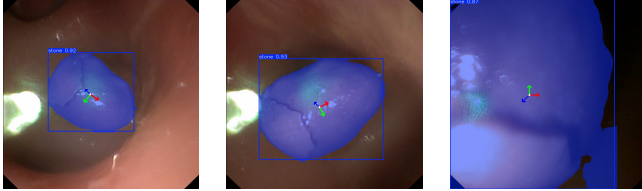
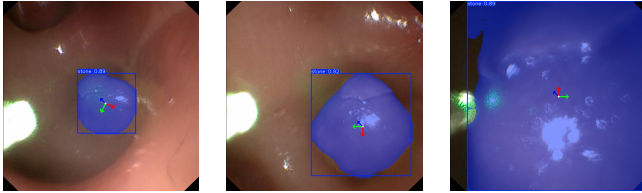


Fig. 3: Examples from our validation set, showing two images for each difficulty category.



(a) Trial 21



(b) Trial 23

Fig. 4: Perception results for two trials: initial scope pose, intermediate pose, and the final ready-to-fire pose.

TABLE III: Segmentation Performance Across Validation Set Difficulties

Model	Easy				Medium				Difficult			
	P	R	mAP50	Dice	P	R	mAP50	Dice	P	R	mAP50	Dice
Full Synthetic	0.79	0.80	0.77	0.14	0.80	0.67	0.68	0.14	0.67	0.45	0.51	0.08
Mixed	0.98	0.95	0.99	0.14	0.95	0.79	0.90	0.23	0.84	0.76	0.86	0.22
Full Real	0.95	0.93	0.98	0.03	0.91	0.81	0.85	0.17	0.79	0.89	0.89	0.10
Lu et al. Model	0.25	0.43	0.27	0.34	0.12	0.26	0.12	0.20	0.11	0.27	0.15	0.21

VI. CONCLUSION AND FUTURE WORK

Our offline segmentation experiments demonstrated that while a purely synthetic dataset establishes a strong baseline, blending a vast amount of synthetic data with a minimal fraction of real clinical images (as little as 5%) yields exceptional results. This mixed training regime achieved a Mask mAP50 of 0.933, effectively matching and slightly outperforming a network trained on a fully real dataset of 1,433 images. Crucially, the mixed model exhibited remarkable resilience across varying clinical difficulty tiers, maintaining high accuracy even when target stones were heavily obscured by procedural artifacts.

Furthermore, we successfully validated the practical utility of our perception pipeline by deploying it on our physical ARA-URS platform. Using a closed-loop visual servoing control scheme governed by our trained YOLO26 instance segmentation model, the robot autonomously navigated to a target stone in a benchtop kidney phantom. The system reliably closed the distance to the target, achieving an average

TABLE IV: Kidney Stone Screen Coverage Results

Screen Coverage Percentage per Trial (%)										
Trial 01	Trial 02	Trial 03	Trial 04	Trial 05	Trial 06	Trial 07	Trial 08	Trial 09	Trial 10	
36.50	74.34	29.28	20.18	22.26	26.51	26.91	20.44	39.64	54.02	
Trial 11	Trial 12	Trial 13	Trial 14	Trial 15	Trial 16	Trial 17	Trial 18	Trial 19	Trial 20	
55.85	100.00	60.50	50.21	51.34	53.87	51.64	100.00	53.63	100.00	
Trial 21	Trial 22	Trial 23	Trial 24	Trial 25	Trial 26	Trial 27	Trial 28	Trial 29	Trial 30	
75.10	100.00	87.35	100.00	50.81	61.38	100.00	46.99	49.04	61.58	
Average Coverage (%): 57.13%							Trials with > 50% Coverage: 20 / 30			

terminal screen coverage of 57.13% and proving the viability of synthetic-to-real transfer for dynamic surgical robotics.

Despite these promising results, a measurable sim-to-real gap remains when relying exclusively on purely synthetic data, particularly in visually degraded conditions. Future work will focus on narrowing this gap by introducing more advanced physically-based modeling of dynamic surgical artifacts, such as floating dust from laser lithotripsy, active bleeding, and fluid turbulence. Additionally, we plan to extend the perception framework to encompass multi-object tracking—including the real-time segmentation of surgical tools like laser fibers, safety wires, and extraction baskets.

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